

Causal Inference with Ordinal Outcomes: Flexible Estimation Using Normalizing Flows

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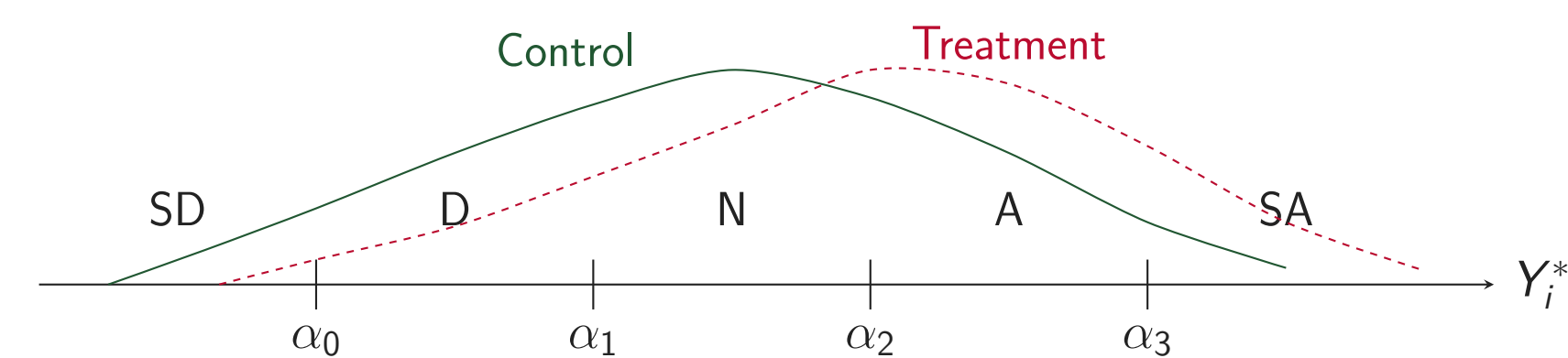
Motivation

Likert responses are ordinal, not cardinal. Estimating mean differences therefore imposes untestable measurement assumptions.

Target causal effects on the observed ordinal response distribution, not coefficients on an arbitrary latent scale.

- ▶ Latent attitudes are observed only through ordered categories.
- ▶ Latent treatment effects are identified only up to location and scale.
- ▶ Category-specific and cumulative probability effects remain directly interpretable.

Ordinal Measurement and Estimands



Treatment shifts the latent distribution across fixed reporting thresholds.

The target object is the distribution of potential ordinal outcomes:

$$\pi_j(d) = P\{Y_i(d) = j\} = \mathbb{E}_X[P(Y_i = j \mid D_i = d, X_i)].$$

- ▶ **Category-specific effects:** $\Delta_j = \pi_j(1) - \pi_j(0)$.
- ▶ **Cumulative effects:** $\Delta_{\geq j} = P\{Y_i(1) \geq j\} - P\{Y_i(0) \geq j\}$.

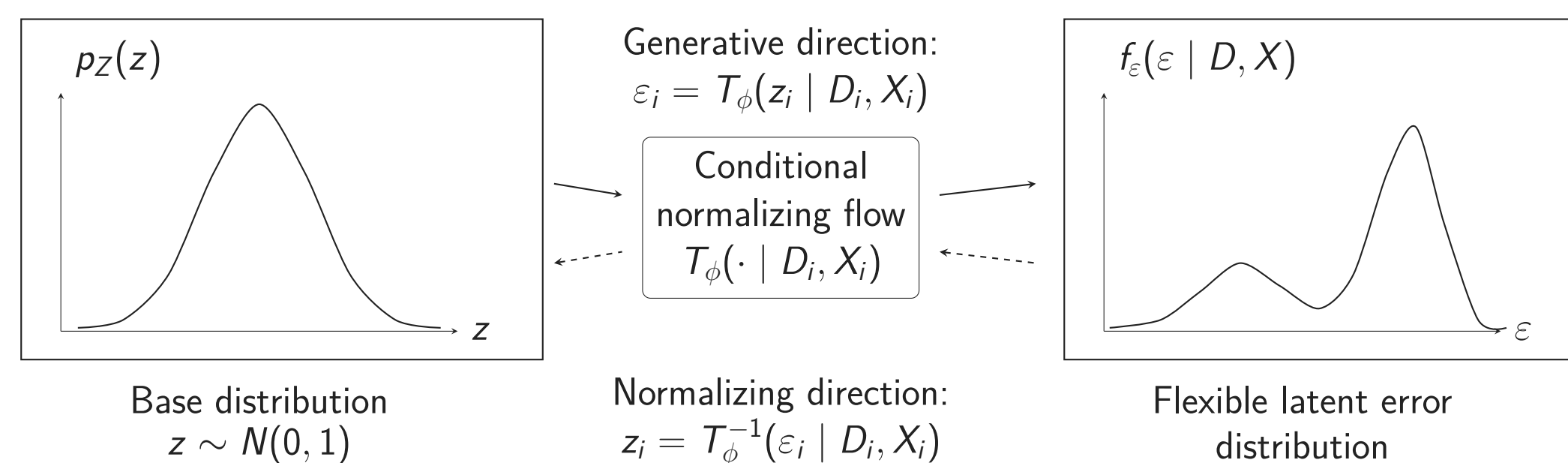
Estimation Problem

- ▶ Ordered logit and ordered probit $\implies$ **fixed latent error distribution** and a **parametric index**.
- ▶ When attitudes are polarized, clustered, skewed, or nonlinear in covariates, these assumptions can be violated.

Approach	Strength	Limitation
OLS	Simple	Assumes cardinal scores
Dichotomization	Easy	Discards ordering info
Ordered models	Respects ordering	Fixed error distribution
Flexible flows	Likelihood-based	Higher computational cost

Latent treatment effects are identified only up to location and scale.

Conditional Normalizing Flows

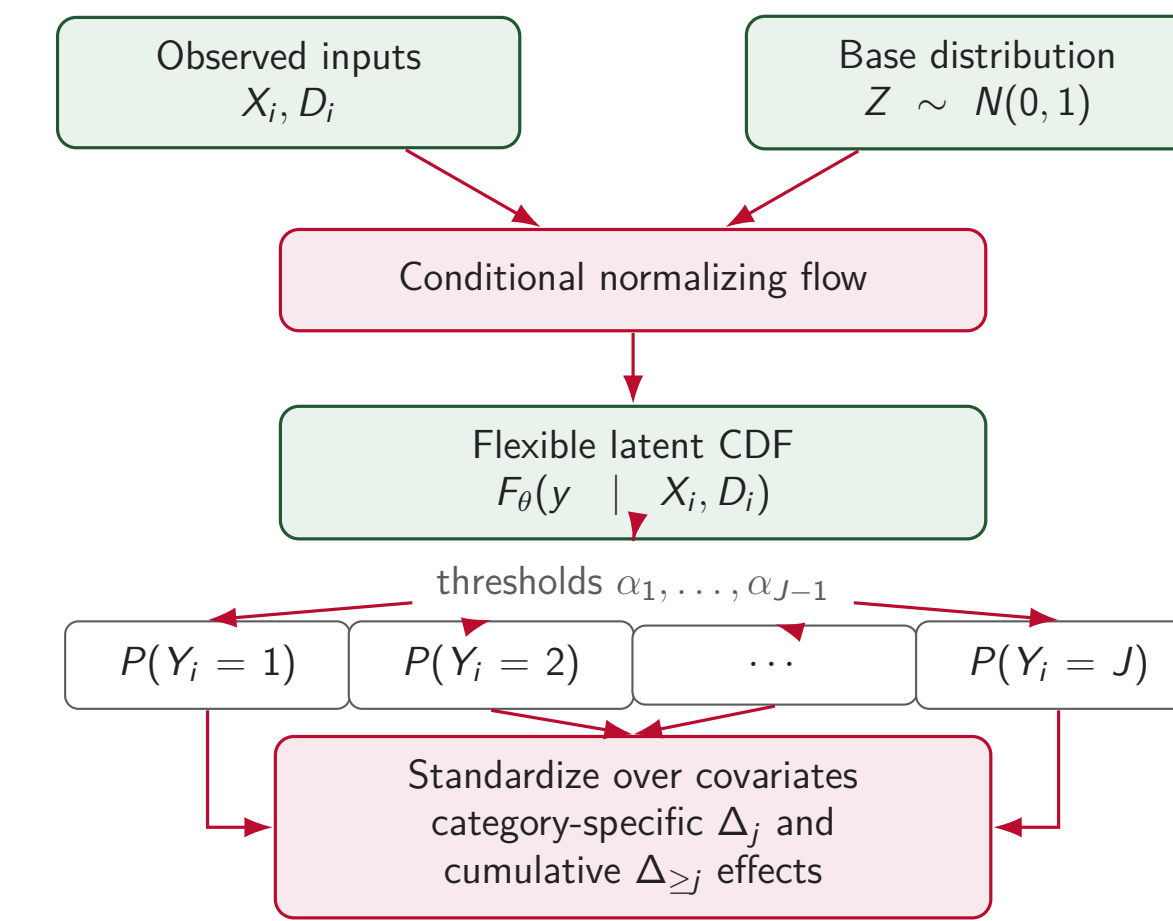


Combined with the latent index and thresholds, the estimated error distribution implies conditional ordinal probabilities $P(Y_i = j \mid D_i, X_i)$.

In the proposed framework, this is constructed using monotonic rational-quadratic splines (RQS), replacing the parametric error family while retaining analytical tractability and exact likelihood evaluation.

Proposed Framework

The estimator keeps the threshold-crossing structure but replaces restrictive parametric assumptions with a conditional normalizing flow.



For each response category,

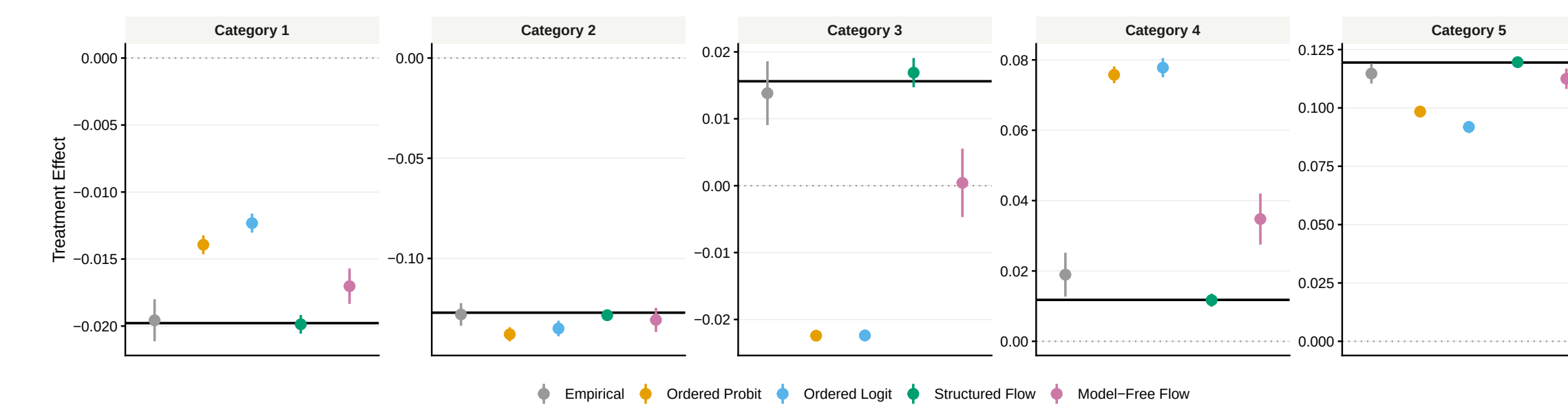
$$P(Y_i = j \mid X_i, D_i) = F_\theta(\alpha_j \mid X_i, D_i) - F_\theta(\alpha_{j-1} \mid X_i, D_i).$$

We estimate $\Theta = \{\tau, h, \phi, \alpha\}$ directly by maximizing the log-likelihood:

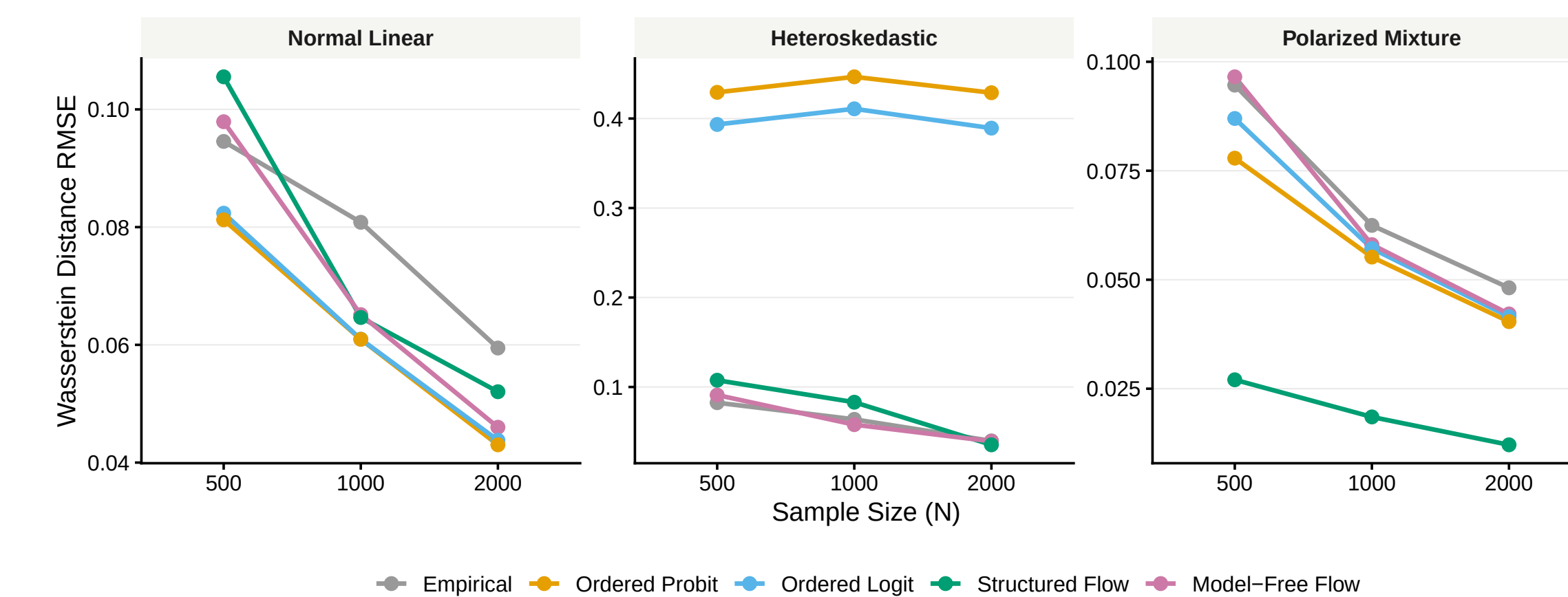
$$\ell(\Theta) = \sum_{i=1}^n \sum_{j=0}^J \mathbb{I}(Y_i = j) \log [F_\theta(\alpha_j \mid X_i, D_i) - F_\theta(\alpha_{j-1} \mid X_i, D_i)]$$

This provides a genuine likelihood rather than a surrogate loss, retaining exact inference.

Monte Carlo Simulation



Single-index model with bimodal error: category-specific treatment effects.



RMSE of Wasserstein distance between estimated and true response distributions.

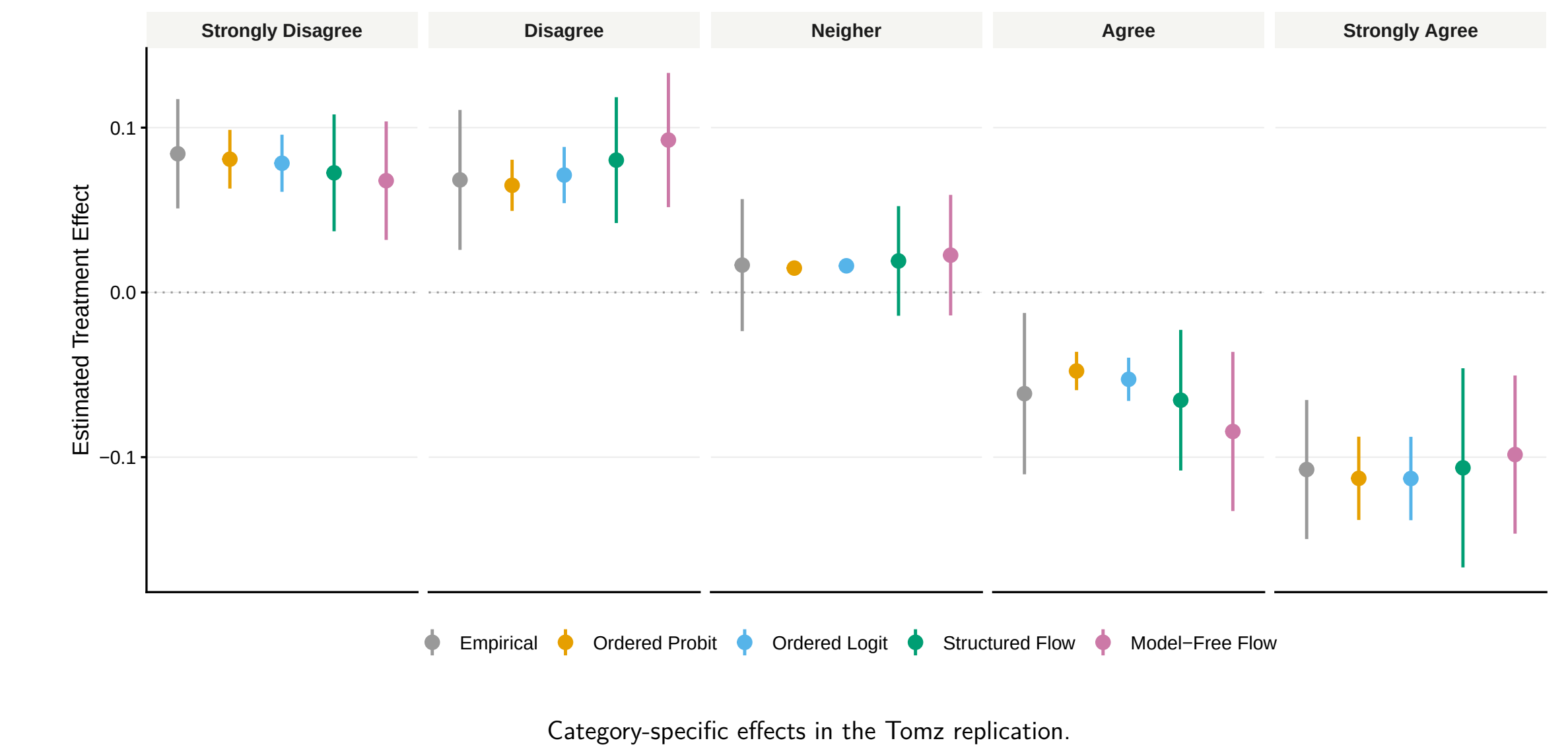
- ▶ Ordered response models perform well when assumptions are approximately correct.
- ▶ When latent-distribution assumptions fail, ordered logit/probit can distort category-level treatment effects.
- ▶ Structured flow estimators improve robustness while retaining statistical efficiency.

Empirical Replications

Two published survey experiments illustrate when flexible estimation confirms conventional findings and when it changes substantive interpretation.

Human rights and support for military intervention (Tomz and Weeks, 2020)

Estimated treatment effects are broadly stable across empirical, ordered response, and flow-based estimators, suggesting conventional models are adequate in this setting.



Category-specific effects in the Tomz replication.

Authoritarian propaganda and public opinion (Mattingly et al. 2025)

Estimated effects differ across modeling assumptions, especially in middle and upper response categories. Flexible distributional modeling changes which categories absorb the treatment-induced probability shift.



Category-specific effects in the Mattingly replication.

Take Away

- ▶ **Report Probability-Based Quantities:** Avoid interpreting raw latent coefficients. Focus on category-specific (Δ_j) and cumulative ($\Delta_{\geq j}$) probability effects.
- ▶ **Assess Distributional Sensitivity:** Use flexible flow estimators as a robustness check to ensure findings are not driven by arbitrary normal or logistic assumptions.

Contact and Materials

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Code & Data: https://github.com/chanhyuk58/ordinal_outcome